

Topic 1: Introduction to Prompt Engineering

1.1 What is Prompt Engineering?

- Prompt engineering is the practice of designing and refining inputs to AI language models to get accurate, useful, and reliable outputs
- A prompt is any text (or combination of text and data) you send to an AI model as input
- The quality of the output is directly shaped by the quality of the prompt -- small changes in wording can dramatically change results
- Prompt engineering is a skill, not magic -- it follows learnable patterns and principles
- Used by: developers building AI applications, analysts querying data with AI, writers, researchers, and educators

1.2 Why Prompt Engineering Matters

- LLMs (Large Language Models) are general-purpose -- they need context and direction to perform specific tasks well
- A vague prompt produces a vague answer; a precise prompt produces a precise answer
- Prompt engineering reduces hallucinations (confident but wrong answers) by constraining the model
- It enables non-programmers to build powerful workflows using plain language
- Well-engineered prompts can replace hundreds of lines of traditional code for certain tasks

1.3 The Prompt Engineering Mindset

- Think of the model as an extremely capable but literal assistant -- it does exactly what you say, not what you mean
- Be explicit: state the task, the format you want, the audience, and any constraints
- Iterate: treat prompt writing like software development -- write, test, observe, refine
- Understand failure modes: models can be overconfident, biased, or context-blind
- Document your prompts: track what works and why, just like code comments

1.4 Applications of Prompt Engineering

- Text generation: summaries, reports, emails, essays, documentation
 - Code generation: writing, explaining, debugging, and translating code
 - Data extraction: pulling structured information from unstructured text
 - Question answering: building chatbots, FAQs, and knowledge assistants
 - Classification and sentiment analysis: labelling, tagging, and categorising text
 - Creative tasks: brainstorming, storytelling, ideation, and design
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